OPERATING SYSTEM

SCHEDULING ALGORITHMS

HARDWARE TROJANS (HT)

- A **hardware trojan** is a malicious modification of an integrated circuit's circuitry, designed to disrupt operations, steal data, or cause physical harm or death.
- These modifications can be inserted during the design or manufacturing stages, often through untrust third-party components or intellectual property cores.

CHARACTERISTICS OF HARDWARE TROJANS

1. **Size** – This refers to the number of components added or modified. Trojans can range from **small transistors** to **large circuits**.
2. **Distribution**

i. **Tight Distribution** – Trojan components are located close together in the layout.

ii. **Loose Distribution** – Components are dispersed across the chip making detection harder

CHARACTERISTICS OF HARDWARE TROJANS (CONTINUES…)

3. **Structure** – Changes in the physical layout can alter the chips (IC) dimensions or component placement.
4. **Type**

i. **Functional Trojans –** These involve adding or deleting components such as transistors or logic gates to modify chip's functionality.

ii. **Parametric Trojans** – These involve modifying existing components such as thinning wires or weakening transistors, to affect the chip's reliability or performance.

IMPACT OF HARDWARE TROJANS

1. Data Breaches and Intellectual Property Theft
2. System Malfunction
3. Performance Degradation
4. Reputational Damage and Financial Costs
5. Denial – of – service (DoS) Attacks

MITIGATION AND REMEDIATION TECHNIQUES OF HT

1. Detection Techniques
2. Collaborative Methods
3. Message Authentication Codes (MACs)

QUESTION

Calculate the average waiting time and average turnaround time. For:

1. First Come First Serve (FCFS) Scheduling Algorithm

FIRST COME FIRST SERVED SCHEDULING

PROCESS ID (PID)	ARRIVAL TIME	BURST TIME
1	4	5
2	6	4
3	0	3
4	6	2
5	5	4

PID	C.T	T.A.T	W.T
1	9	5	0
2	17	11	7
3	3	3	0
4	19	13	11
5	13	8	4

Average = 8 4.4 units

T.A.T = C.T – ARRIVAL TIME

W.T = T.A.T – BURST TIME

C.T = Completion Time

T.A.T = Turn Around Time

W.T = Waiting Time

Gantt Chart

TIME	0	3	-	-	-	4	9	13	17	19
PID		3	Idle time				1	5	2	4

Calculations

0 + 3 = 3

4 + 5 = 9

9 + 4 = 13

13 + 4 = 17

17 + 2 = 19

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FIRST COME FIRST SERVED SCHEDULING (CONTINUES…)

- **Convoy Effect**

If process with higher burst time arrived before the processes with smaller burst time, then, smaller processes have to wait for a long time for longer processes to release the CPU.

SHORTEST JOB FIRST – NON PRE-EMPTIVE

PROCESS ID (PID)	ARRIVAL TIME	BURST TIME
0	0	7
1	2	4
2	4	1
3	5	4

PID	C.T	T.A.T	W.T
0	7	7	0
1	12	10	6
2	8	4	3
3	16	11	7

Average = 8 4 units

T.A.T = C.T – ARRIVAL TIME

W.T = T.A.T – BURST TIME

Gantt Chart

TIME	0	7	8	12	16
PID		0	2	1	3

Calculations

0 + 7 = 7

7 + 1= 8

8 + 4 = 12

12 + 4 = 16

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SHORTEST JOB FIRST – PRE-EMPTIVE

PROCESS ID (PID)	ARRIVAL TIME	BURST TIME
0	0	7 – 2 = 5
1	2	4 – 2 = 2
2	4	1
3	5	4

PID	C.T	T.A.T	W.T
0	16	16	7
1	7	5	1
2	5	1	0
3	11	6	2

Average = 7 2.5 units

Gantt Chart

TIME	0	2	4	5	7	11	16
PID		0	1	2	1	3	0

Read Queue

TIME	5	2	4
PID	0	1	3

T.A.T = C.T – ARRIVAL TIME

W.T = T.A.T – BURST TIME

PRIORITY SHEDULING ALGORITHM

Consider the lowest number to be the highest priority

PRIORITIVE	PROCESS ID (PID)	ARRIVAL TIME	BURST TIME
3	0	0	7
2	1	2	4
4	2	4	1
1	3	5	4

PID	C.T	T.A.T	W.T
0	7	7	0
1	15	13	9
2	16	12	11
3	11	6	2

Average = 9.5 5.5 units

Gantt Chart

TIME	0	7	11	15	16
PID		0	3	1	2

T.A.T = C.T – ARRIVAL TIME

W.T = T.A.T – BURST TIME

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